# A Fast Algorithm for Generating All Triangulations of Biconnected Plane Graphs

Tawkir Aziz Rahman

CSE, BUET

# Overview

# Introduction

**The Problem:**

- Generate **all possible triangulations** of biconnected plane graphs
- Previous methods fail with complex structures due to multi-edge creation
- Important applications in graphics, engineering, and chip design

**Our Contribution:**

- Novel algorithm with **constant time per triangulation**
- Solves the multi-edge problem using **safe root** concept
- Practical for real-world applications

# Algorithm

There will be a lot of different methods in this algorithm

- find all triangulations
- find all child triangulations
- find safe root vertex
- checking if the new chord will be valid

# Find All Child Triangulations

**procedure** FIND-ALL-CHILD-TRIANGULATIONS-CYCLE($T$)
    Output $T$
    **if** $T$ has no generating chords **then**
        **return**
    **end if**
    Let $(v_b, v_{b'})$ be the leftmost blocking chord of $T$
    **for** $j \geq b$ **do**
        **if** $(v_1, v_j)$ is a chord of $T$ **then**
            FIND-ALL-CHILD-TRIANGULATIONS-CYCLE($T(v_1, v_j)$)
        **end if**
    **end for**
**end procedure**

# Find All Triangulations Cycle

**procedure** FIND-ALL-TRIANGULATIONS-CYCLE($n$)
    Output root $T_r$
    $T = T_r$
    **for** $j = n - 1$ **to** 3 **do**
        FIND-ALL-CHILD-TRIANGULATIONS-CYCLE($T(v_1, v_j)$)
    **end for**
**end procedure**

# For all

**procedure** FIND-ALL-CHILD-TRIANGULATIONS$(T, i)$

  Let $E_G$ be the set of generating edges of $T(F_i)$
  **if** $E_G$ is empty **then**
    **return**
  **end if**
  **for all** $e \in E_G$ **do**
    Flip $e$ to find a new triangulation $T'$
    Output $T'$

    **for** $j = 1$ **to** $i$ **do**
      FIND-ALL-CHILD-TRIANGULATIONS$(T', j)$
    **end for**
  **end for**
**end procedure**

# continue

**procedure** FIND-ALL-TRIANGULATIONS$(G, k)$

  Label the faces $F_1, F_2, \cdots, F_k$ arbitrarily
  Find root triangulation $T_R$ of $\mathcal{T}(G)$
  Output *root* $T_R$

  **for** $i = 1$ **to** $k$ **do**
    FIND-ALL-CHILD-TRIANGULATIONS$(T_R, i)$
  **end for**
**end procedure**